Ridhi Bansal
Electromechanical specialist
J&JMedTech



Ridhi Bansal

Johnson&Johnson
MedTech

Orthopaedics

Role and work

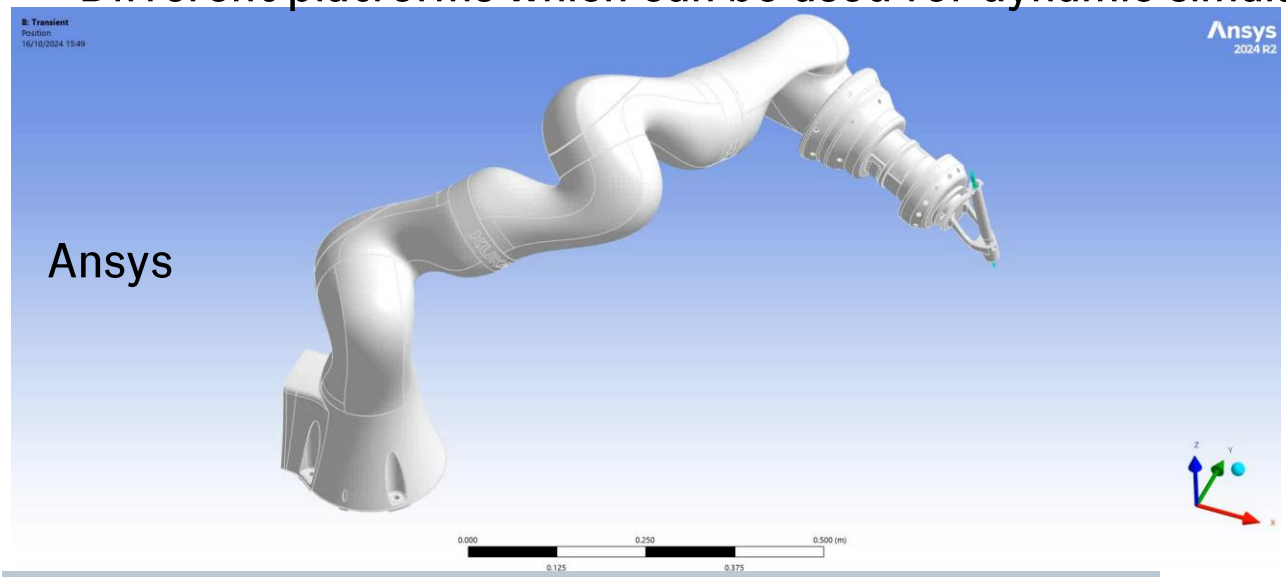
- Creating simulation models–
  - Reduces testing on Physical setup
  - Help to create parts (suggest part designs based on simulation results)
  - Validation and verification
  - Some areas physical testing can be completely replaced by simulation testing.

- The work is demand and priority-based and customer focused: receive requests from R&D to analyse any product in the development pipeline (implants and instruments).
- The work is multidisciplinary and spans the full J&J Orthopaedics business including joints (knee, hip, shoulder), trauma, spine and digital/capital.
- Understand what is required and accordingly collaborate with team/stakeholders to deliver the results.
- Manage project and scope. Manage expectations and any additional requests are directed to subsequent requests.
- In the digital/capital space, the role is to act as a conduit between software and hardware teams to generate new models.
- Collaborate with wider teams across different departments and work with international teams.

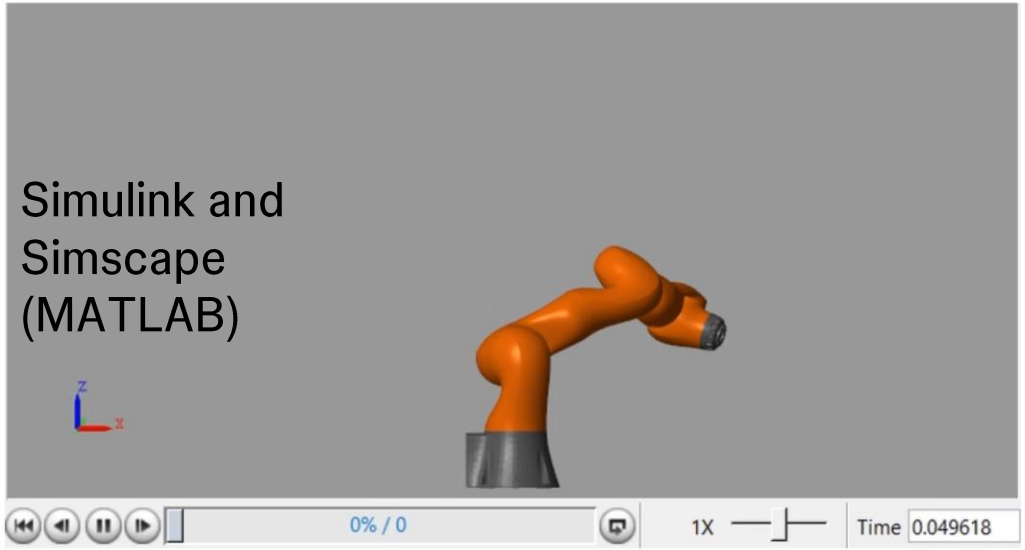
Projects

Aspen (Velys) Spine Robot

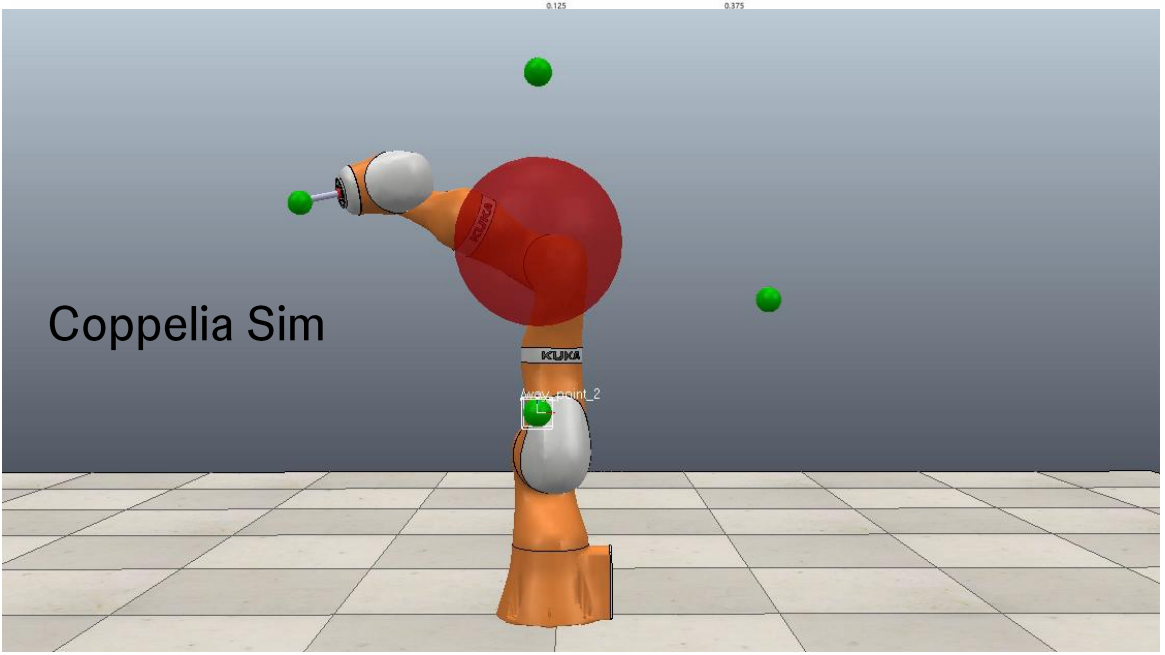
Different platforms which can be used for dynamic simulation of KUKA motion.



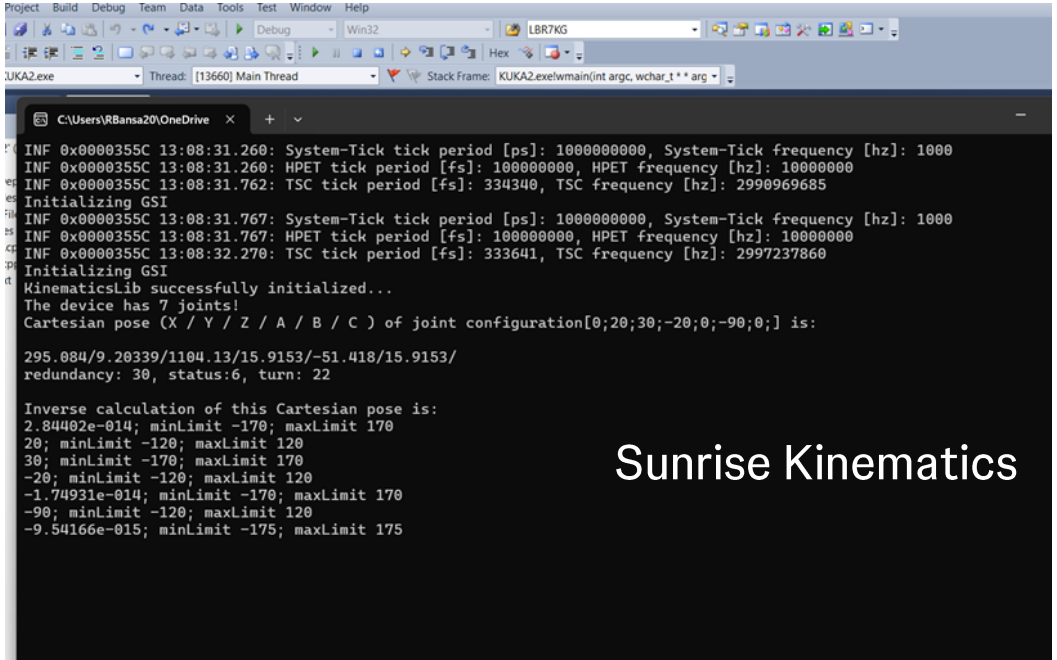
Ansys



Simulink and Simscape (MATLAB)

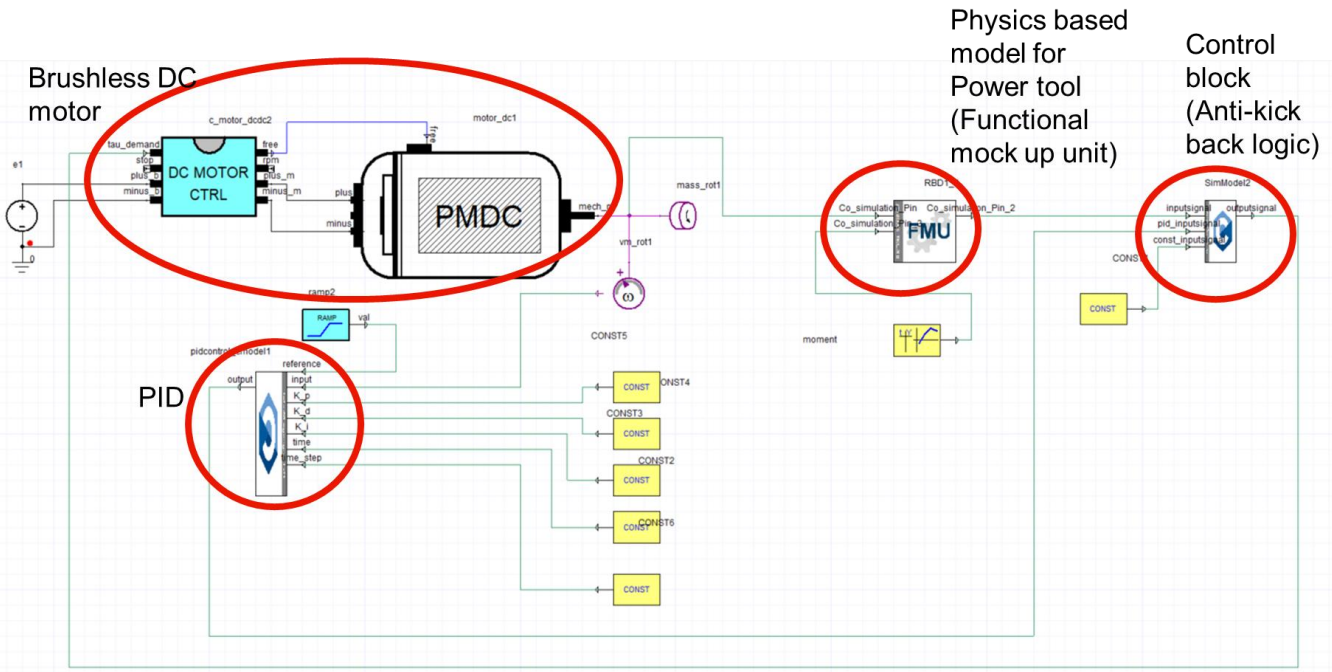


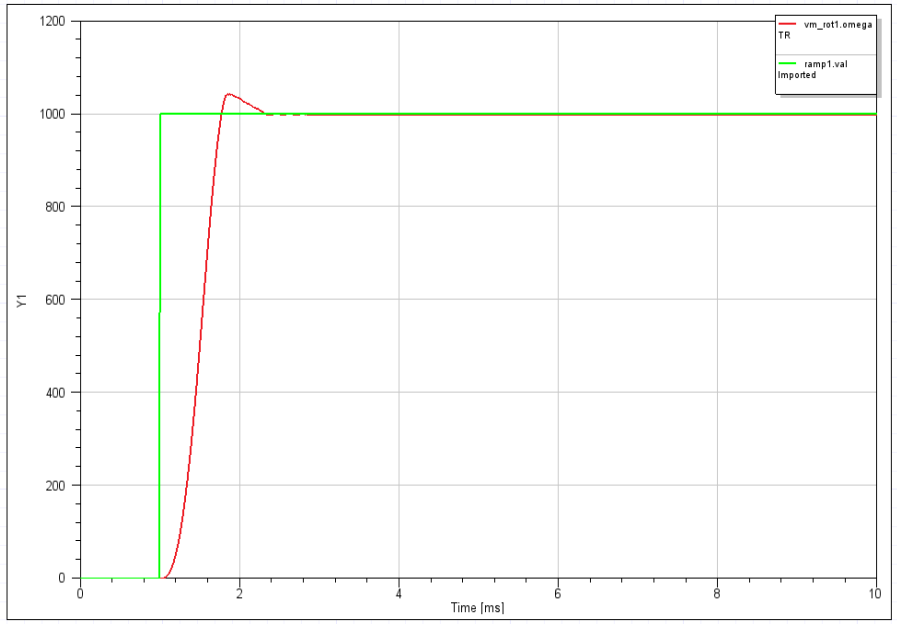
Coppelia Sim

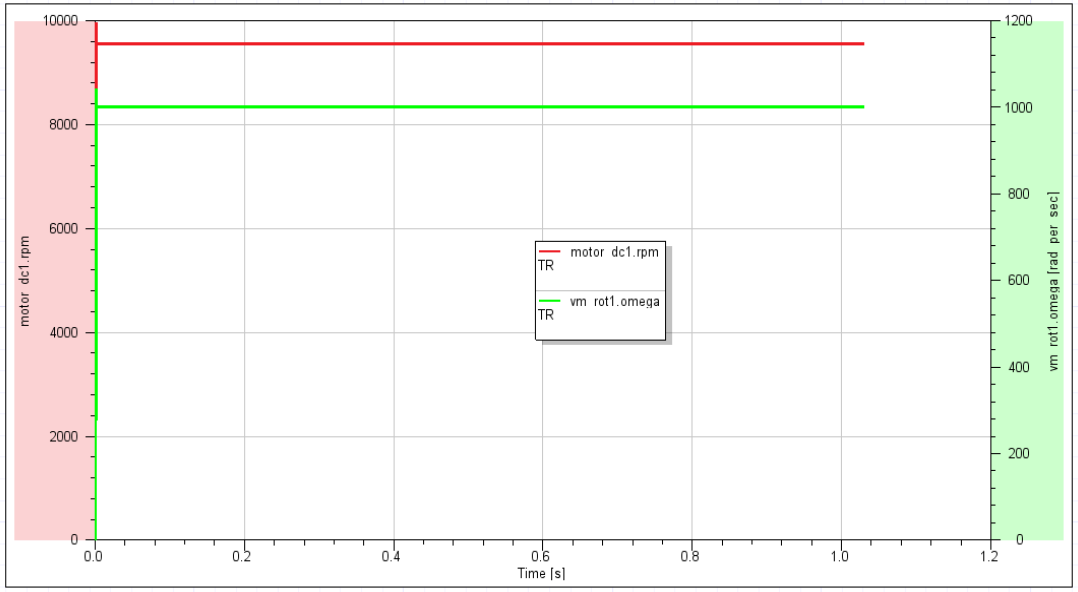


Sunrise Kinematics

Apollo - Powertool





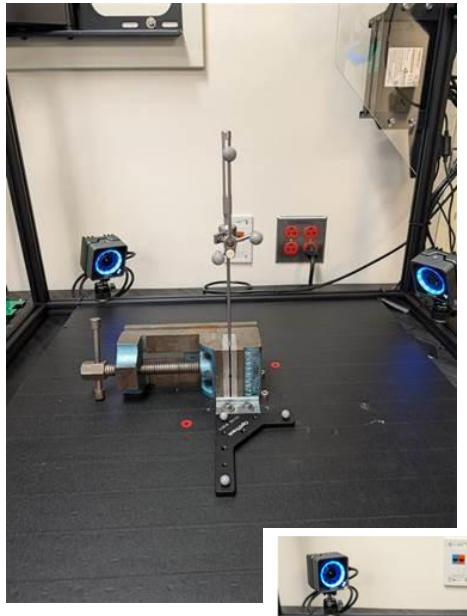


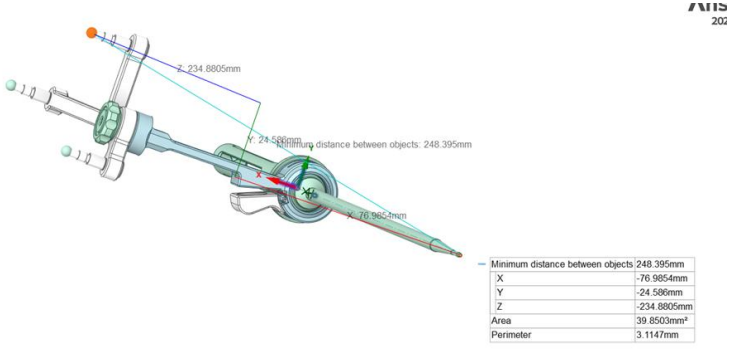
Create a simple system model of the Powertool and motor to input signal and get the output vibrations which can be monitored to control the motor.

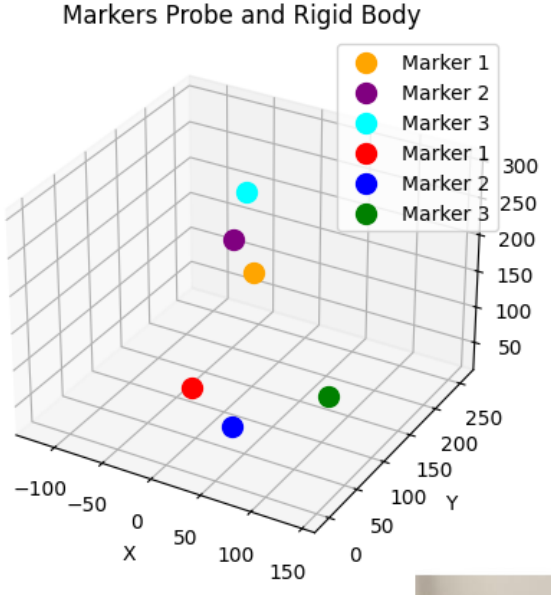
This is a new business area which is expanding on simulation testing. Challenge : To understand what challenges the team would encounter during the running of drill and design the model accordingly.

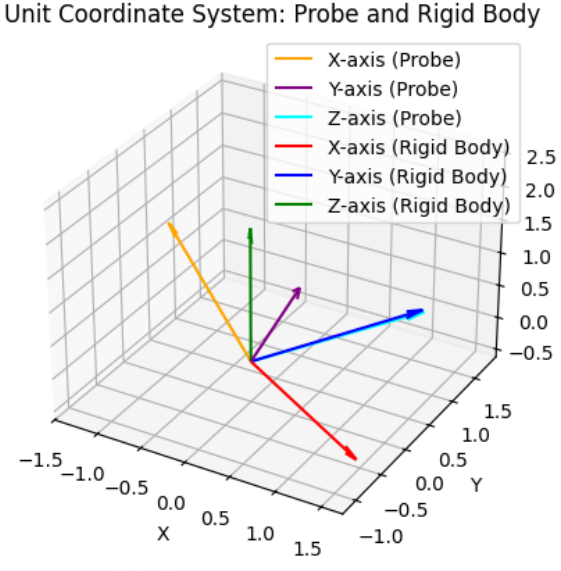
Optitrack – Aspen robot tracking robot movement

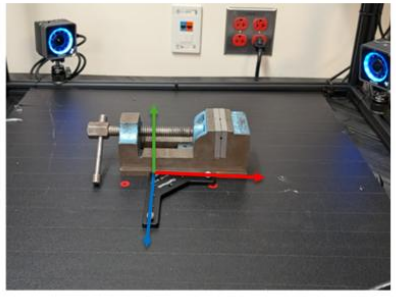
This method provides the user with a consistent test method to develop a navigation accuracy test method that can help in evaluation of positional accuracy of test parts - using the Optitrack System. The aim is to measure the angular position of any instrument with markers on it and the point-to-point distance between this instrument with respect to any other point in space to determine the percentage positional accuracy of the Optitrack system.
Developed code to analyse the raw data from the system and conduct accuracy testing.

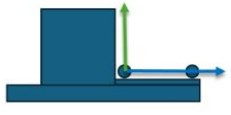


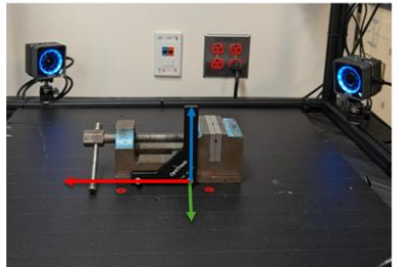


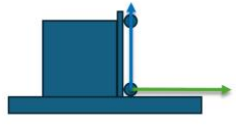








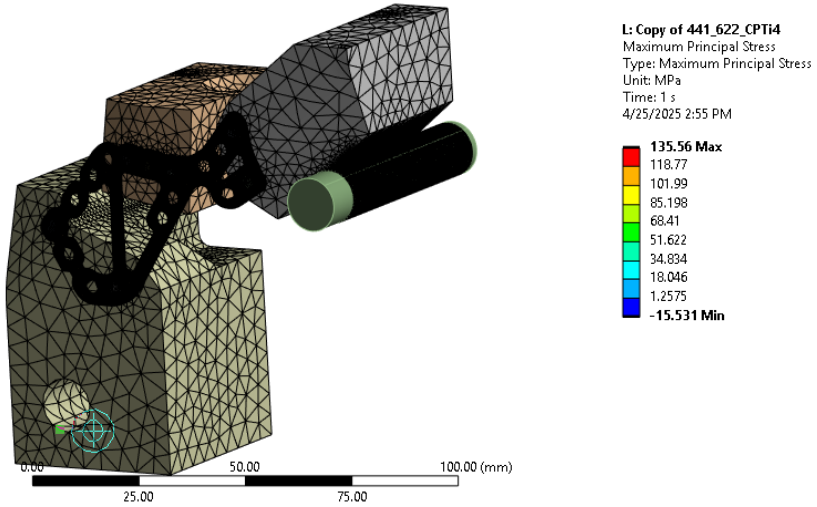


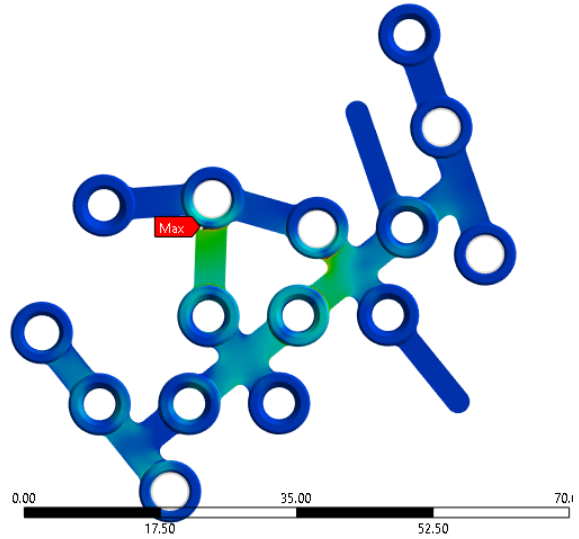


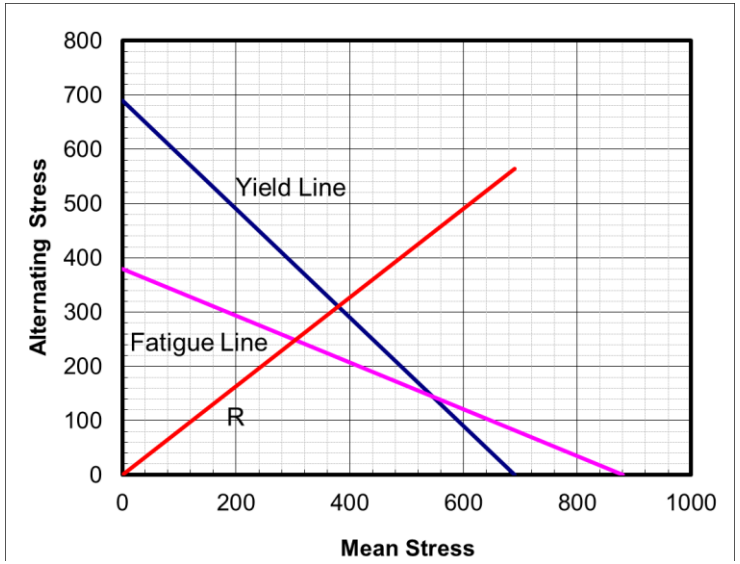
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Ankle Trauma Plate

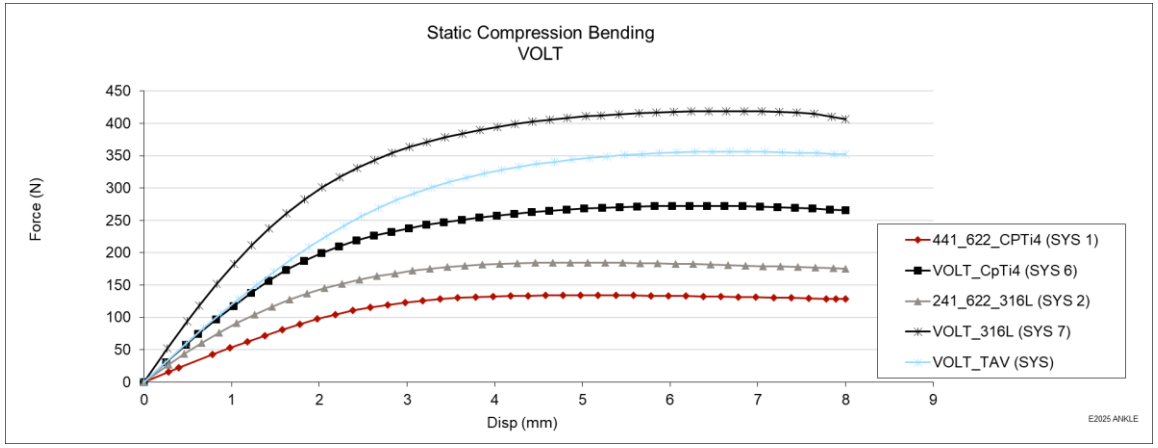
The calcaneal plate represents a significant advancement in orthopedic technology, providing essential support for the management of calcaneal fractures. To do Yield and fatigue study for the angle plates under loading conditions. The study was done for 5 different plates (2 different designs and 3 different materials).







Goodman Diagram – SS 316L

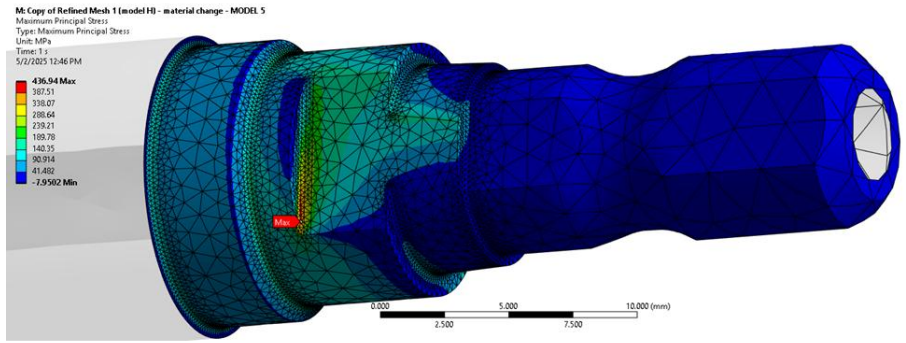


Plates	Max Principal stress under 10 N load	Maximum Allowable Stress (from Goodman Diagram)	Fatigue Load Maximum Loading Force (Load for the Maximum Allowable stress condition
	(MPa)	(N)	(N)
441_622_CPTi4 (SYS 1)	135.56	433	10 × 433 / 135.56 = 31.94
VOLT_CpTi4 (SYS 6)	81.955	433	10 × 433 / 81.955 = 52.83
VOLT_CpTi4 (SYS 6) (excluding region of holes)	64.23	433	10 × 433 / 64.23 = 67.41
241_622_316L (SYS 2)	119.44	552	10 × 552 / 119.44 = 46.21
VOLT_316L (SYS 7)	70.7	552	10 × 552 / 70.7 = 78.07
VOLT_TAV (SYS)	64.409	692	10 × 692 / 64.409 = 107.43

Voyager

Project Voyager is a shoulder navigation project based on novel electromagnetic tracking technology. To do Yield and fatigue study for the driver shaft when Torque is applied to it.
The study was done for driver shafts made with different martials.

- Created different models for different Torque values.
- Created different models for different mesh size and refinements.



Model Number	Material	Global mesh sizing (mm)	Sizing on edges and stepped face (mm)	Sizing on stress face edge	Element number Total Model
1	SS 316 L	2	0.2	N/A	221679
2	SS 316 L	1.5	0.15	N/A	321265
3	SS 316 L	2	0.15	0.15	297648
4	SS 316 L	3	0.3	0.3	184758
5	Ti6Al4V - wrought/forged [a+b]	2	0.2	N/A	221679

Model number	Maximum Stress (Equivalent (von-Mises)) (MPa) for 0.2 mm fillet radius	Maximum Stress (Equivalent (von-Mises)) (MPa) for 0.5 mm fillet radius	Maximum Stress (Max principal) (MPa) for 0.2 mm fillet radius	Maximum Stress (Max principal) (MPa) for 0.5 mm fillet radius	Material	Yield Strength, min (MPa)	Tensile Strength, min (MPa)	Allowable stress (From Goodman Diagram) (MPa)	Fatigue strength/Endurance limit
1	479.19	460.73	441.99	332.5	SS 316 L	550 [1]	795 [1]	530	50% UTS: $795 \times \frac{50}{100}$ = 397.5 MPa
5	480.13	464.29	436.94	330.17	Ti6Al4V	795 [2], 860 [3]	860 [2], 930 [3]	573 [for 2], 620 [for 3]	50% UTS: $860 \times \frac{50}{100}$ = 430 Mpa [for 2], $930 \times \frac{50}{100}$ = 465 MPa [for 3]



Material Inputs	
Yield Strength	550
UTS	795
RBF	397.5
Test Inputs	
R Value	1.00E+10

	YS Line	RBF/UTS
σa	275	265
σm	275	265
Fatigue	550	530
Allowable	530	

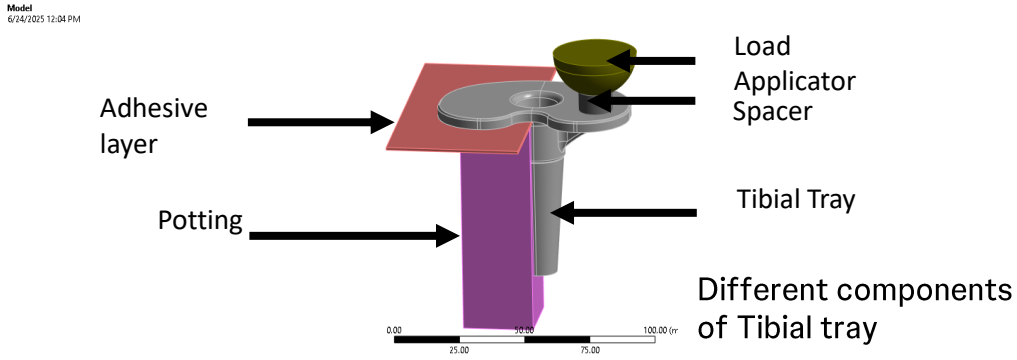
Attune Thick base

GxP Work

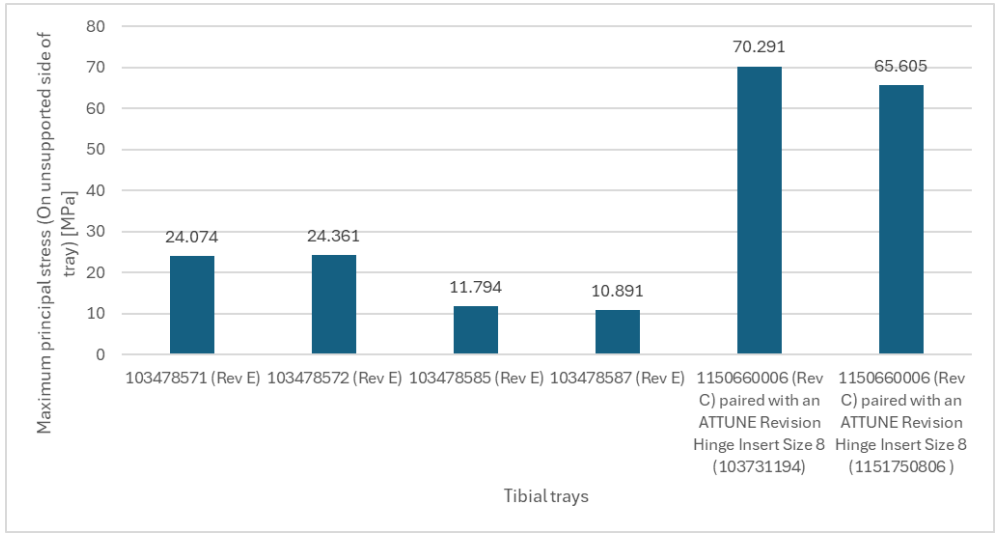
The ATTUNE Revision Rotating Platform (RP) Thick Tibial Bases (Figure 1) are part of the ATTUNE Revision Knee System. The ATTUNE Revision RP Thick Tibial Bases allow the ability to address revision scenarios where significant proximal tibia bone loss has occurred while maintaining ATTUNE Revision compatibility. The RP Thick Tibial Bases provide customers with an alternative method of managing flexion and extension gaps during revision knee arthroplasty via tibial base implants with additional thicknesses. Compare ATTUNE Revision RP Thick Tibial Bases with predicate to do verification & validation study

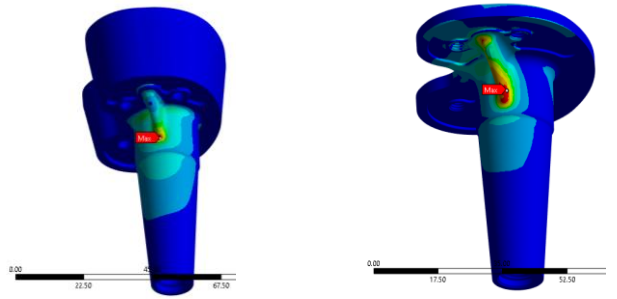
		Model form		Medium
Validation	Computational Model	Model Inputs	Quantification of Sensitivities	Medium
			Quantification of Uncertainties	Medium
	Comparator	Test samples	Quantity of Test Samples	Low
			Range of Characteristics of Test Samples	Low
			Measurements of Test Samples	Low
			Uncertainty of Test Sample Measurements	Low
		Test conditions	Quantity of Test Conditions	Low
			Range of Test Conditions	Low
			Measurement of Test Conditions	Low
			Uncertainty of Test Conditions Measurements	Low
	Assessment	Equivalency of input parameters		Low
		Output comparison	Quantity	Low
			Equivalency of Output Parameters	Low
			Rigor of Output Comparison	Low
			Agreement of Output Comparison	Low
Applicability		Relevance of quantities of interest		High
		Relevance of the validation activities to the COU		Medium

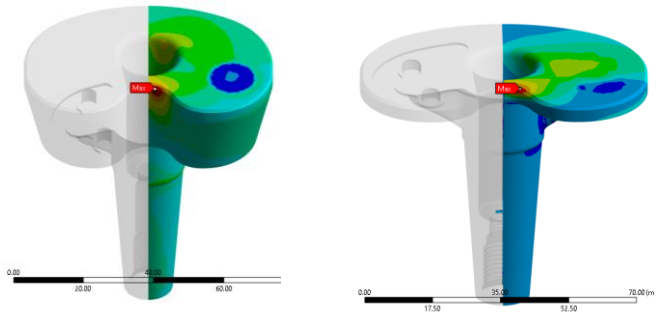
Risk Assessment



Different components of Tibial tray







Stress comparison (FEA)

Transformation work

Hip simulators

To complete onboarding of hip-simulators.

Work instruction and risk analysis for hip simulators

Risk Assessment and SSW Ref: 00

Site		**Department**	
Task/Process		**Assessor(s)**	
Photo		**Safety Signs**	

Likelihood of Occurrence (L)		**x** **Severity (S)**		**=** **Risk (R)**	
Very likely	5	Fatality	5	1-4	Low = L
Likely	4	Severe injury	4		
Fairly likely	3	Lost time injury	3	5-9	Medium = M
Unlikely	2	Minor injury	2		
Very Unlikely	1	No injury	1	10-25	High = H

No.	**Hazard** What could hurt you?	**Persons Affected**	**Existing controls**	**Initial Risk Score** (with current controls)		**Further controls required**	**Residual Risk Score** (after further controls)	
				L x S	**=R**		**L x S**	**=R**
1.	Temperature	Operator	Temperature in the Laboratory is controlled.	1x1	1	No further action required	N/A	N/A
2.	Floor	Operator	Housekeeping and cleaning.	1x3	3	No further action required	N/A	N/A



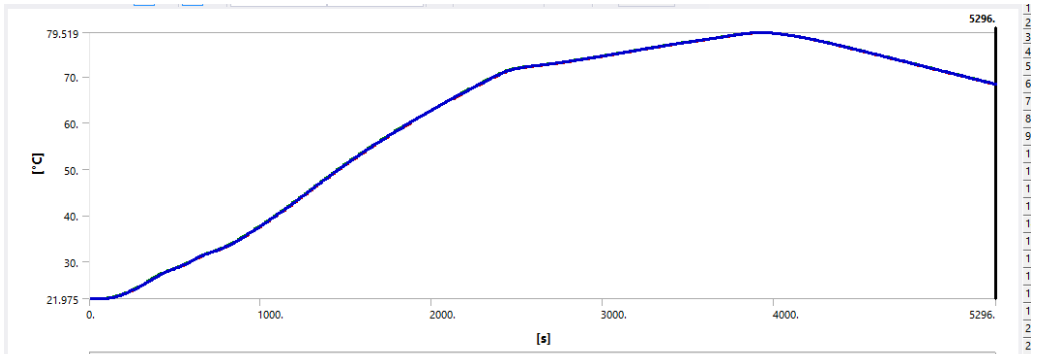
Apollo Heat transfer

Heat transfer in autoclave

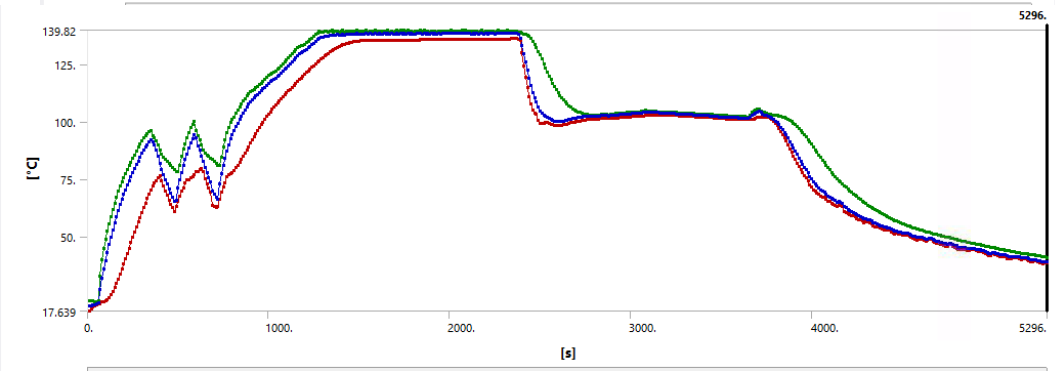
For an autoclave cycle see the change in temperature change with time at internal surface of enclosure and at cell.

- Look at conduction and convection through different surfaces to see the heat transfer.

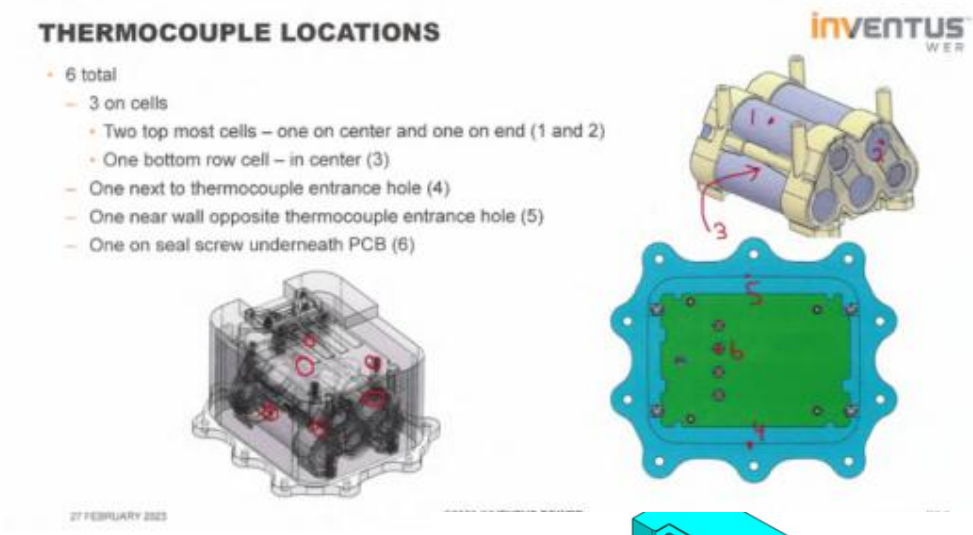
Model input temperature vs time for an autoclave cycle
Material: Enclosure –Polyethylene, Cells –Aluminium.
Air modelled as multi-body part.

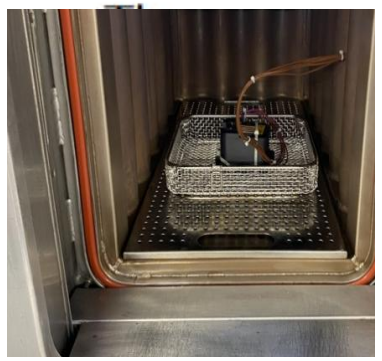


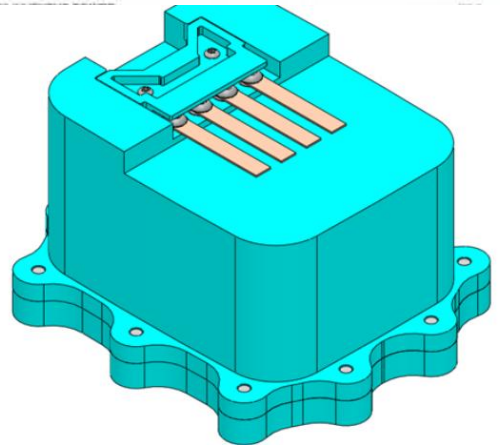
Cell surface temperature



Inner enclosure surface temperature





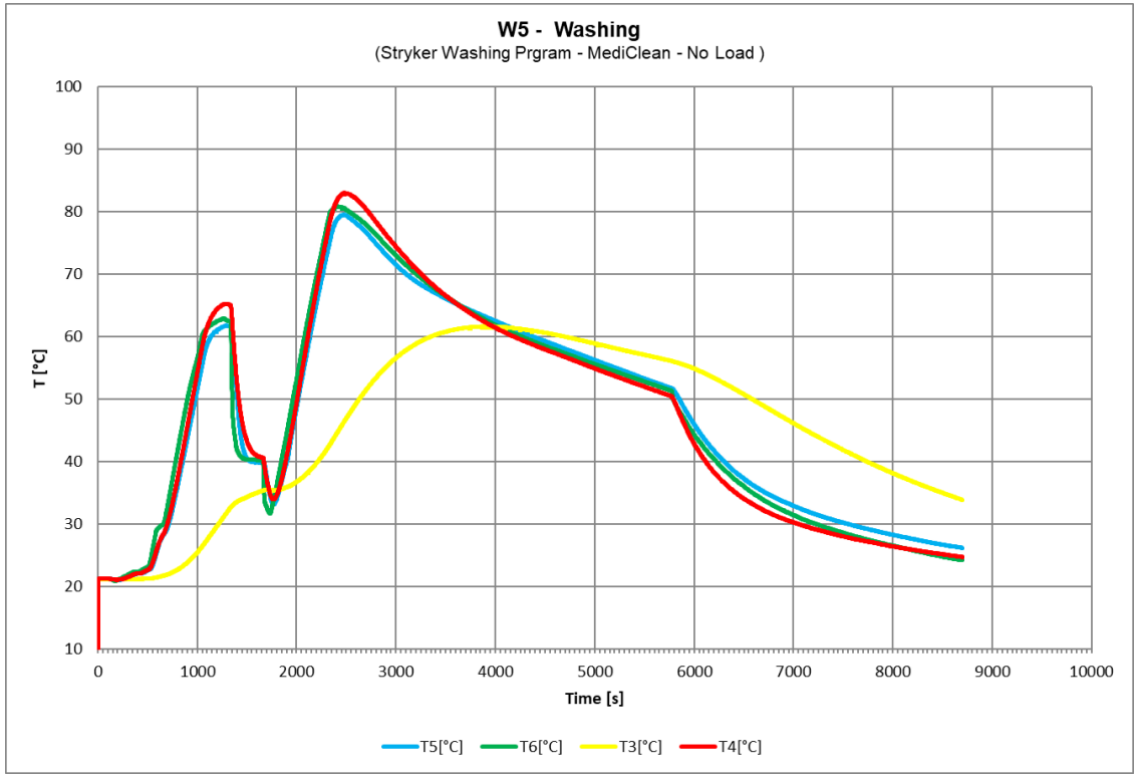


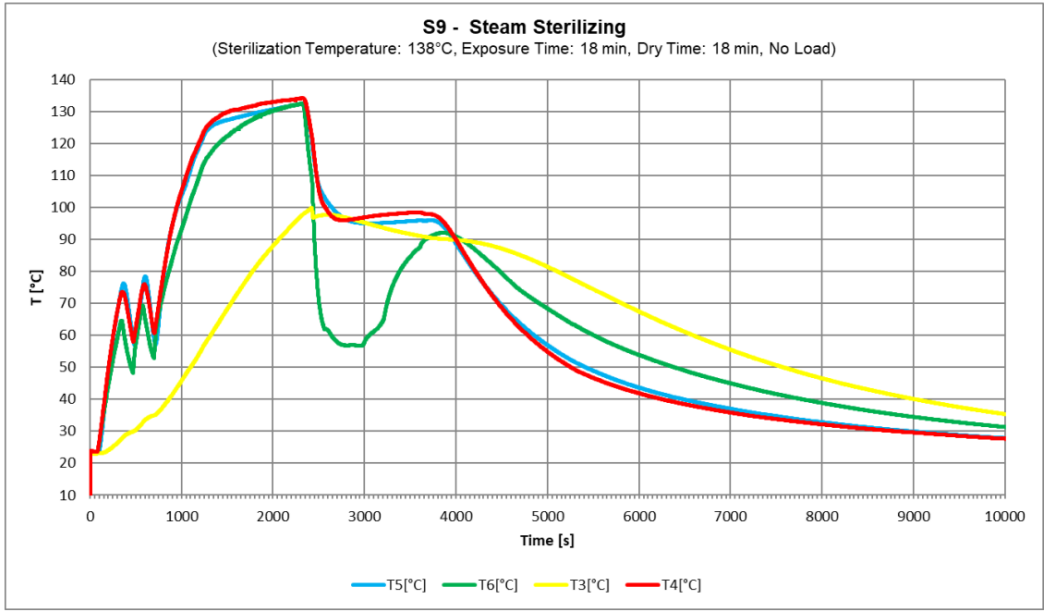
Apollo Heat transfer

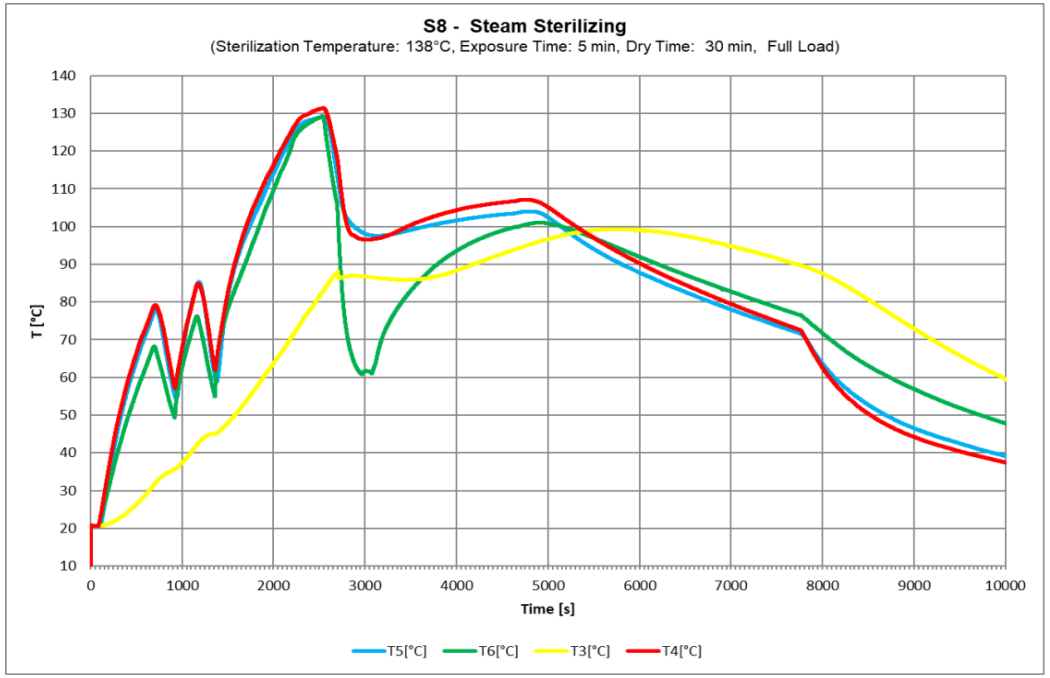
Things to consider

Material of enclosure and cells

Effect of steam due to seal being compromised (green line drop in graph in autoclave whereas in washing cycle all tied up together).







Other projects

Things to consider

- Discovery- CFD analysis, heat analysis, fan, fluid flow
- Drop test
- Develop these models cfd, drop test v&V
- Code validation
- Urgent request support + build of solutions (RnD things which would help in future (pick up by talking to team)
- 6 sigma Green belt trained
- Instron trained, shadow work+ lab work- mechanical/manufacturing

Project Challenges and Deliverables

Project Challenges and Deliverables

- For different projects use different software – Ansys (SpaceClaim, Mechanical, Electronics, Workbench), Coppelia, Sunrise Kinematics, Simulink, SimScape, MATLAB, Minitab, VS code etc.
- Different challenges in projects:
  - Compatibility between different software.
  - Communication between two software with embedded models can be challenging.
  - Defining correct Boundary and Constraint Conditions is crucial.
  - Robot joints need to be defined properly.
  - Experimental setup can affect the error calculation results between simulation and physical experiments.
  - Calibrated tools/ CAD models with exact dimensions and precise set-up dimensions required to avoid errors.
  - Complexity geometries (e.g. threads, sharp edges, screw heads, etc.) in CAD models for FEA analysis need to be removed and meshing on different regions need to defined properly (e.g. high stress regions, contact areas, fillets, etc.).
- Based on the obtained results suggestions are proposed and then the simulations are executed again with changes in set-up or model.